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Torsional behaviour of horizontal joints in prestressed concrete towers for wind turbines

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Abstract: The horizontal joint is a critical part in the prestressed concrete towers for wind turbines; however, the design methods in existing codes fail to accurately predict its torsional capacity under combined bending and shear actions. This study aims to improve the design accuracy of the horizontal joint in prestressed concrete towers for wind turbines. An experimental investigation was conducted on four horizontal joint specimens subjected to combined compression-torsion or compression-bending-shear-torsion loads. The results indicated that the torsional capacity of horizontal joints was primarily attributed to the friction between the upper and lower segments, and the torsional capacity of horizontal joints was significantly influenced by the bending moment and shear force applied to the horizontal joint. Furthermore, a finite element (FE) model was developed and validated against the experimental results, and a parametric study was conducted to investigate the influence of key parameters including the material strengths, cross-sectional dimensions, prestressing force and transverse loads. Subsequently, the applicability of the existing design methods of torsional capacity for horizontal joints was evaluated and found to be inadequate due to their neglect of the bending moment and shear force. Finally, a modified method is proposed and validated through the experimental and numerical results obtained in this study. Results demonstrate that the proposed design method is more accurate and consistent for predicting the torsional capacity of horizontal joints in prestressed concrete towers for wind turbines.

Keyword: Bending-torsion behaviour, Concrete tower, Horizontal joint, Prestressing force, Wind turbine.

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1 Introduction

The development of renewable wind energy is crucial for addressing the current energy crisis. Steel-concrete hybrid structural towers offer high lateral stiffness and low construction and maintenance costs, making them the most popular type of towers for onshore wind turbines with heights exceeding 140 meters [1-6]. The prestressed concrete towers for wind turbines are typically constructed using prefabricated concrete ring segments through a modular assembly method. As shown in Fig. 1, the concrete shells are connected to one another using arc-shaped bolts and epoxy resin to form the ring segment, which is referred to as the vertical joint. The ring segments are connected using epoxy resin, and the connection joints between the ring segments are referred to as horizontal joints (see Fig. 1). This new type of connection not only enhances construction efficiency but also facilitates levelling [7-8].

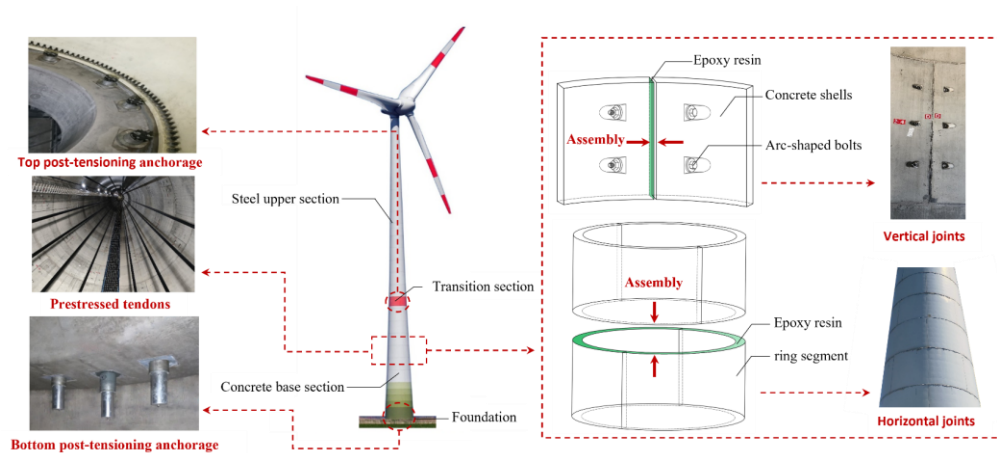


Fig. 1 Schematic drawing of the steel-concrete hybrid tower

Under the wind load and prestressing force, the prestressed concrete towers for wind turbines are subjected to the combined compression, bending, shear and torsion, with the bending moment being the governing action. Stress redistribution occurs when the bending moment acts on the hybrid towers, particularly at weak connections of horizontal joints, which will affect the damaged area of the joint. At ultimate, cracks may develop at the horizontal joints, indicating that conventional section design methods for thin-walled tubular structures are not applicable for the torsional design of these joints [9-11].

Recently, several studies have examined the applicability of current design codes for predicting

the torsional capacity of horizontal joints in prestressed concrete towers for wind turbines. Paredes et al. [12] reported that the presence of connection joints between ring segments does not influence the overall structural stiffness; however, it significantly affects the extent of structural damage, as indicated by numerical simulations. Hartwig and Middendorf [13] conducted detailed theoretical analysis and numerical simulation of the concrete segments in prestressed concrete towers subjected to combined bending, shear and torsion. They found that the torsional capacity of horizontal joints is maximized under pure torsion conditions; however, it is significantly affected by the bending moment and shear force applied to these joints. Loh and Garg [14-16] identified deficiencies in the existing design methods for assessing the torsional capacity of horizontal joints in prestressed concrete towers through theoretical analysis. They proposed a new calculation model for the torsional capacity of these joints under various cracking conditions, and provided calculation parameters for different cracking scenarios. The design codes [17] employ the Saint-Venant equation to calculate the torsional capacity of horizontal joints. This equation is derived from the equivalent calculation of the ultimate torque of the thin-walled hollow section tube. However, the equivalent calculation of the torsional capacity has been found to be overly conservative [18-19]. Kang et al. [20] presented a detailed derivation of the torsional calculation theory for horizontal joints in existing design methods, specifically *DIN EN 1992-1-1* [17]. They utilized finite element (FE) results to demonstrate the over-conservativeness of the design method for predicting the torsional capacity of horizontal joints after cracking. In addition to the aforementioned methods, the calculation method proposed by the classical mechanical model for thin-walled circular hollow section beams under combined shear and torsion [21] employed the superposition of torsional and shear loads and assumed that the cross-section of the member is continuous, ignoring the movement of the cross-sectional torsional centre after cracking, which can significantly affect the overall performance of the tower.

In the aforementioned investigations, the existing design methods for the torsional capacity of horizontal joints revealed several issues: (a) When the horizontal joint is uncracked but subjected to transverse loads, the interface friction of the joint is the only factor resisting torque and shear forces. However, the influence of shear forces on the torsional capacity of horizontal joints remains unclear. (b) After the horizontal joint cracks, the torsional properties of the joint change, and the torsional centre of the section shifts, which directly affects the calculation of the torsional capacity of

horizontal joints. However, this aspect has not been considered in any of the existing design methods.

Therefore, this study aims to develop more accurate torsional design methods for the horizontal joints in prestressed concrete towers for wind turbines. This study begins with an experimental program involving four large-scale horizontal joints. Next, a FE model is developed and validated against the test results. Subsequently, a parametric study is conducted to generate a total of 126 numerical results, investigating key parameters such as material strength, cross-sectional dimensions, prestressing force, and transverse loads. Based on the newly generated experimental and numerical results, the existing design methods are evaluated, and a new design method is proposed.

2 Experimental programme

2.1 Specimens

In the experimental programme, a total of four specimens were tested, as summarised in Table 1. Each specimen was assembled from a lower segment, an upper segment, and a transition segment. The prestressing force (F_{py}) (= 195 kN) of the specimens was designed based on the actual prestressed concrete towers for wind turbines. The average compressive strength of the concrete was measured to be 82.3 MPa, while the average yielding and ultimate stresses of the HRB400 steel reinforcement were found to be 447 MPa and 560 MPa, respectively.

Table 1 Test parameters of specimens

Specimen number	Load scenario.	Wall thickness (mm)	Transverse loads (kN)
PCT-1	Pure torsion	75	0
PCT-M-1	Bending-torsion	75	75
PCT-M-2	Bending-torsion	75	90
PCT-M-3	Bending-torsion	90	105

2.2 Test setup

The height and diameter of specimens were 3000 mm and 650 mm, respectively. The horizontal joint was situated between the upper and lower segments. The specimen was loaded using two 100T hydraulic servo actuators coherently, as shown in Fig. 2. The upper and lower segments were connected by eight prestressed tendons inside the specimen. The prestressed tendons sequentially passed through the loading connection device, transition segment, upper segment, lower segment, and foundation connecting the specimen from top to bottom. The ends of the prestressed tendons were anchored at the top of the loading connection device and at the bottom of the foundation and

an anchor cable pressure sensor was installed at the foundation to monitor the prestressing force of the tendons. The 100T hydraulic servo actuators was connected to the loading connection device and the connection plate using high-strength bolts.

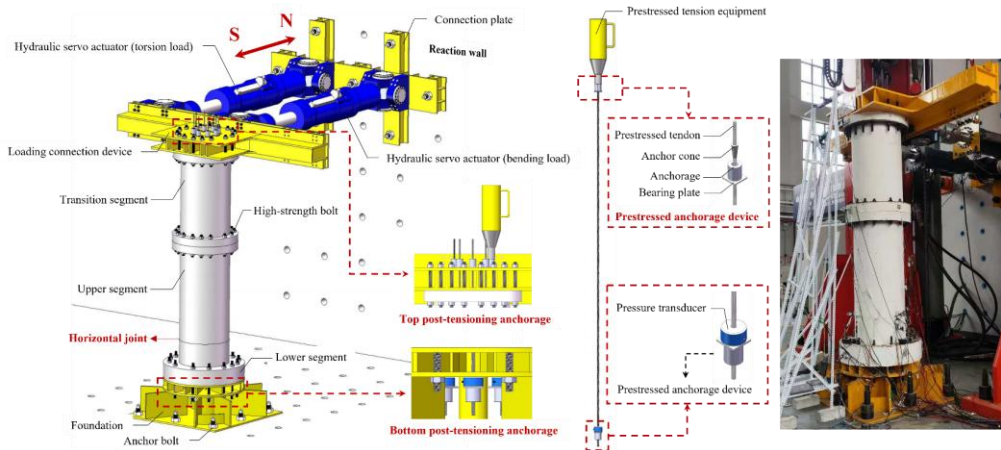


Fig. 2 Test setup

2.3 Measurement scheme and test procedures

During the test, the evolution of strain measurements of the concrete and the vertical displacement of the horizontal joint was recorded at eight locations, as shown in Fig. 3.

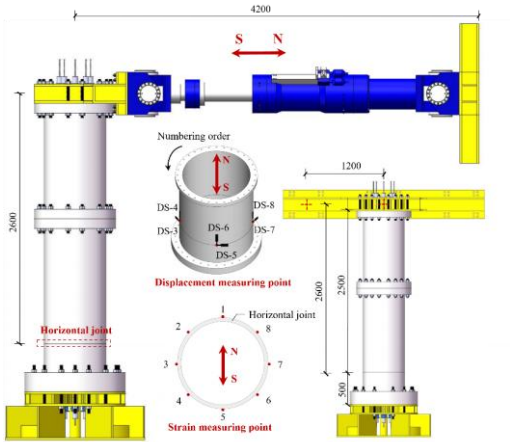


Fig. 3 Displacement and strain testing points (unit : mm)

In this test, a load-displacement dual control method was employed based on the actual force pattern of the hybrid tower. In the preloading stage, a load of 20 kN was applied to verify that the instrument readings were normal. During the loading stage, the procedure began with a loading rate of 0.2 m/s. The loading will pause for every 2-mm displacement; all measured parameters (such as displacement, loads, strain) will be recorded. When the experimental data had been accurately recorded, and then unloaded to 0 kN. The test was stopped when the shear resistance of the specimen was reduced to less than 85%.

3 Test results analysis

3.1 Failure modes

Specimens subjected to pure torsion and combined bending-torsion exhibited pronounced torsional cracks on the concrete surface, as shown in Figs. 4-7. Under pure torsion, the failure was extensive, leading to significant torsional concrete crushing of the lower segment. This observation is consistent with the theoretical assumption that the entire cross-sectional frictional resistance counteracts the applied external torque, as shown in Fig. 4. In contrast, under combined bending-shear-torsion loading, a composite failure mode of torsion and bending was observed. After the formation of distinct torsional cracks, the concrete of specimens in the compression side (the north side) was crushed and the area where the concrete is crushed only appeared on the compression side of the specimen, which also shows that the transverse load will obviously affect the failure range of the specimen, as shown in Figs. 5-7. This is consistent with the conclusions drawn by Paredes et al [12]. Meanwhile, the number of concrete cracks on the tensile side (the south side) of the specimen decreased significantly. This outcome is primarily attributed to the influence of transverse loads on the stress state at the horizontal joint. This is primarily due to the horizontal joint are subjected to the combined bending, shear under the transverse load. Consequently, the concrete material at the horizontal joint of specimens is subjected to a combined compressive-shear stress state, and the friction at the horizontal joint must resist both the external torque and the shear force. As the transverse load increase, this phenomenon is observed that the horizontal joint cracking, altering its stress mode and consequently reducing the torsional capacity of horizontal joints compared to that under pure torsion. This outcome is primarily attributed to the contact area of the horizontal joint of specimen diminishes, leading to greater compressive stress on the compression side of the contact zone under the influence of the bending moment.

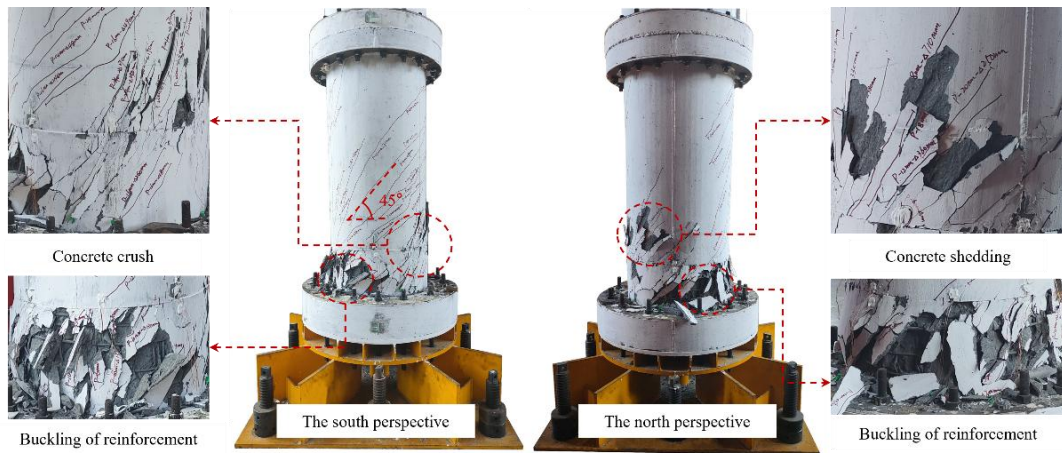


Fig. 4 Failure modes of specimen PCT-1

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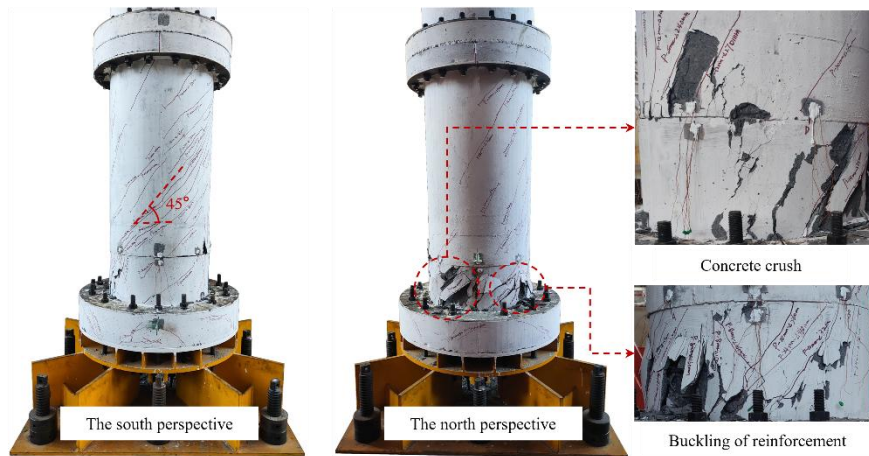


Fig. 5 Failure modes of specimen PCT-M-1

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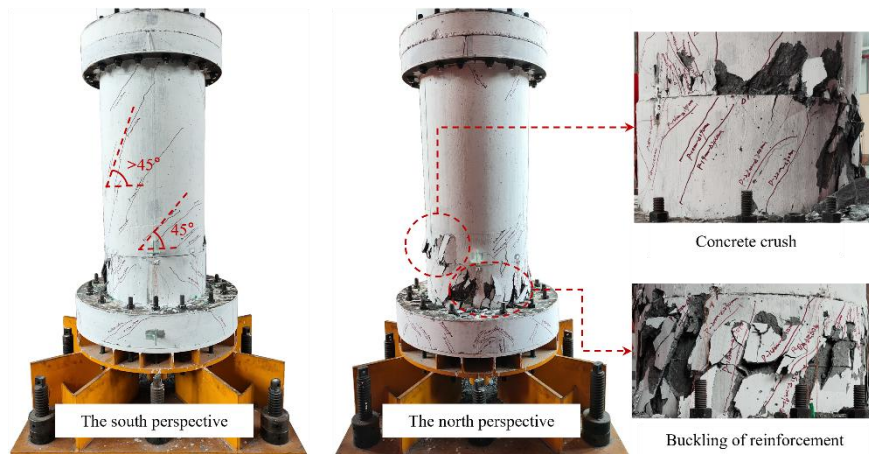


Fig. 6 Failure modes of specimen PCT-M-2

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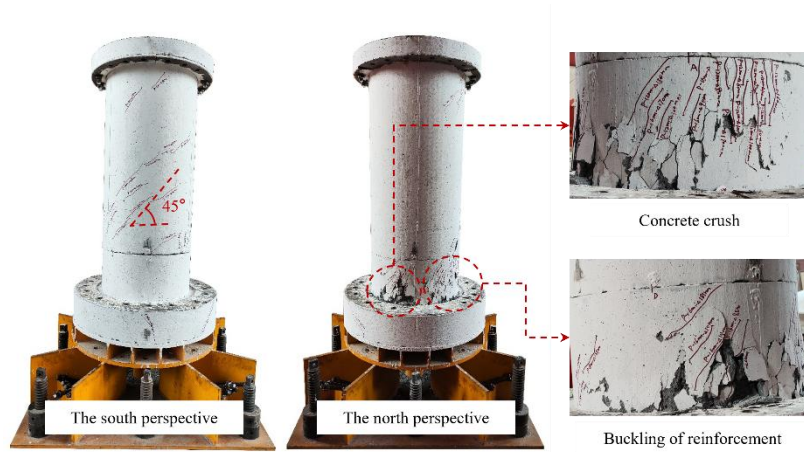


Fig. 7 Failure modes of specimen PCT-M-3

3.2 Torsion angle-torque curve

The torsion angle-torque curves ($T-\theta$) and skeleton curves of specimens are illustrated in Fig. 8. All four specimens exhibited a sudden drop in load after reaching peak values, indicating a brittle failure mode. Meanwhile, the pure torsion specimen demonstrated a higher torsional capacity and stiffness compared to those subjected to combined bending and torsion loads. This difference is primarily attributed to the effects of the bending moment and shear force induced by transverse loads on the overall structural stiffness. As the transverse load increases, the overall stiffness of the specimen decreases. Additionally, the horizontal shear force is countered by friction at the horizontal joint, while the bending moment alters the section properties of the horizontal joint, resulting in a shift of the torsional centre. This shift reduces the effective lever arm of the torque, thereby diminishing the resistance of the torsional capacity. Consequently, as transverse loads increase, the torsional capacity of the horizontal joints decreases. As shown in Figs. 8 (c)-(d), the torsional capacity of the horizontal joints (PCT-M-2 and PCT-M-3) are similar, when cracking occurs in the horizontal joints, it is observed that the torsional capacity of specimens with larger wall thickness does not increase significantly under the same conditions of prestressing force and friction coefficient. This indicates that there is no direct relationship between the torsional capacity of the specimen and the thickness of the section wall. Furthermore, this conclusion that the Saint-Venant equation is unreasonable to calculate the torsional capacity of horizontal joints in prestressed concrete towers for wind turbines was also reported by literature [18-19].

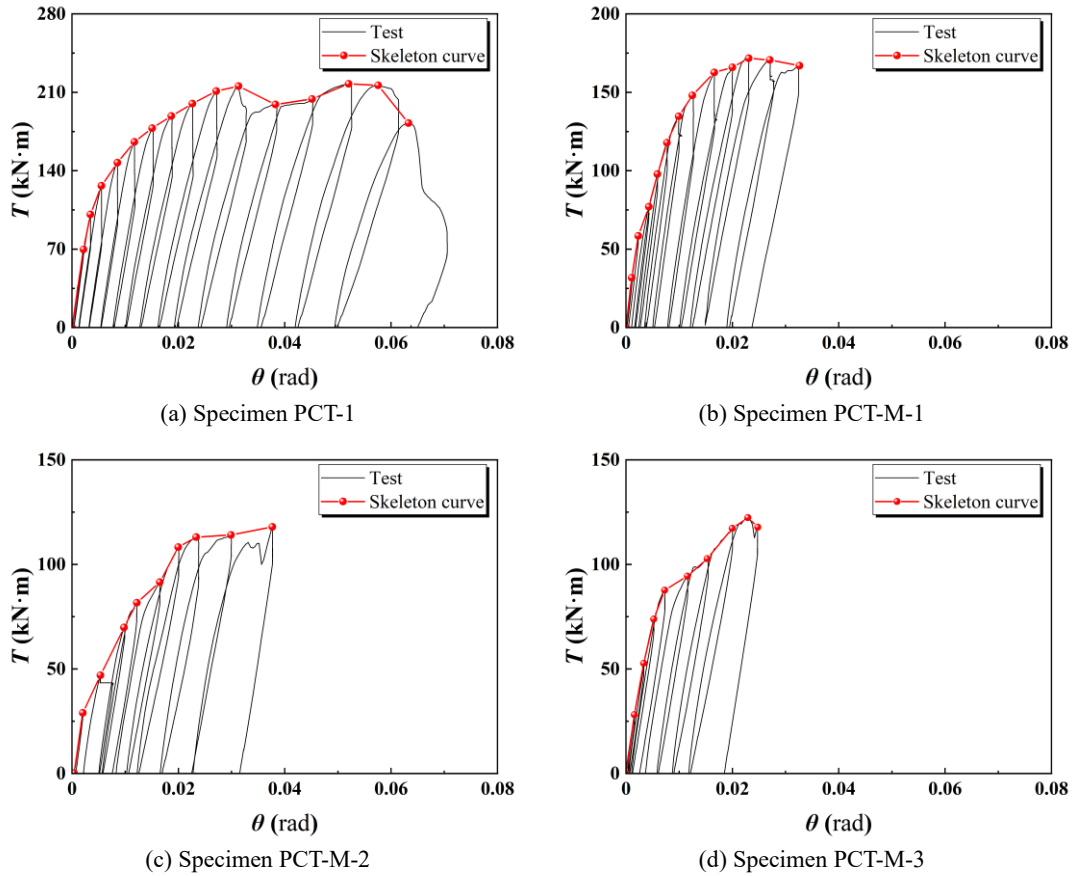


Fig. 8 Torsion angle-torque curves

3.3 Prestressing force increment curve

The torsional capacity of horizontal joints is provided by the friction force at the joint. Therefore, the magnitude of the specimen's prestressing force is the primary factor influencing its torsional behaviour. The trend of the overall prestressing force increment during the loading process is illustrated in Fig. 9. It is evident that the prestressing force of the specimen remains largely constant throughout the loading. However, when the specimen reaches its torsional capacity, the overall prestressing force of specimen slightly decreases due to concrete damage, cracking, and minor localized failure.

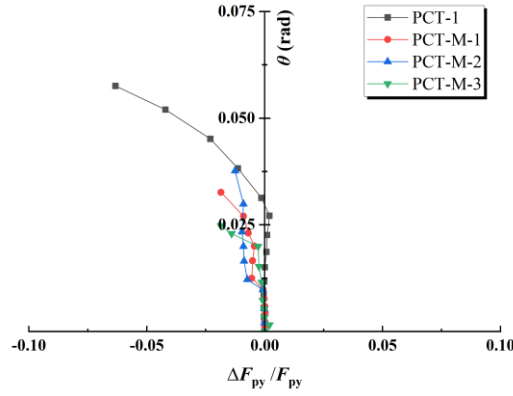
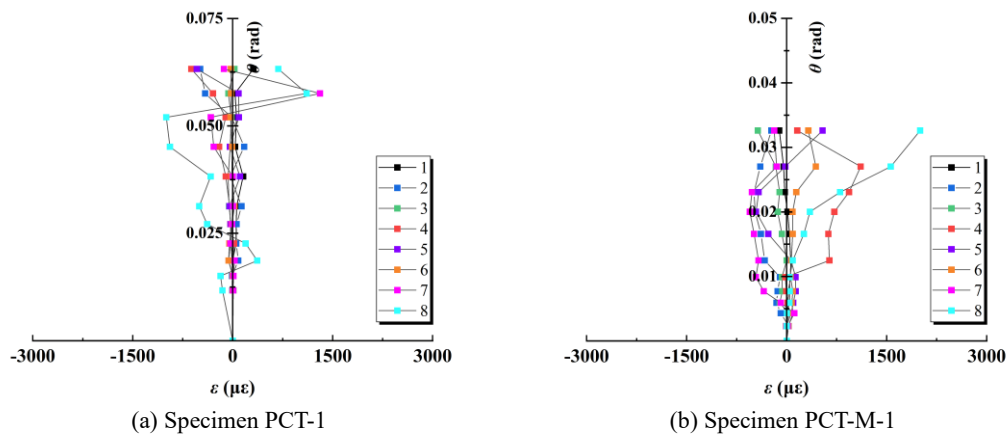


Fig. 9 Prestressing force increment curve

3.4 Strain-displacement curve

This section evaluates the stress distribution and changes in the horizontal joint during loading by extracting peak strain increments vertically along the horizontal joint at each stage, as shown in Fig. 10. The overall vertical strain of the specimen remains essentially constant under pure torsional load. The strain is primarily generated by the vertical prestressing force of the specimen, which aligns with the observed trend of actual prestressing force increment changes. Additionally, the vertical displacement of the eighth measuring point of the specimen PCT-1 exhibits significant positive and negative alternating fluctuations. This phenomenon is primarily attributed to structural damage occurring first at this location, which ultimately becomes the concentrated failure area of the specimen. It is evident that the initial strain of the specimens is relatively stable, as shown in Figs. 10(b)-(d). Notably, the overall trend of strain change exhibits some individual variations at measurement point eight for specimen PCT-1 and measurement point seven for specimen PCT-M-2. These variations are likely due to the horizontal joint not being perfectly flat, which leads to stress concentration and structural damage at those specific locations.



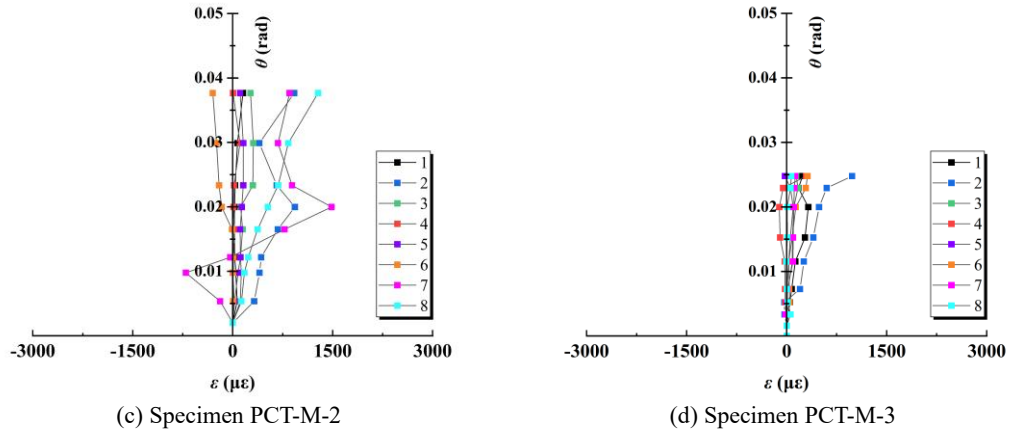
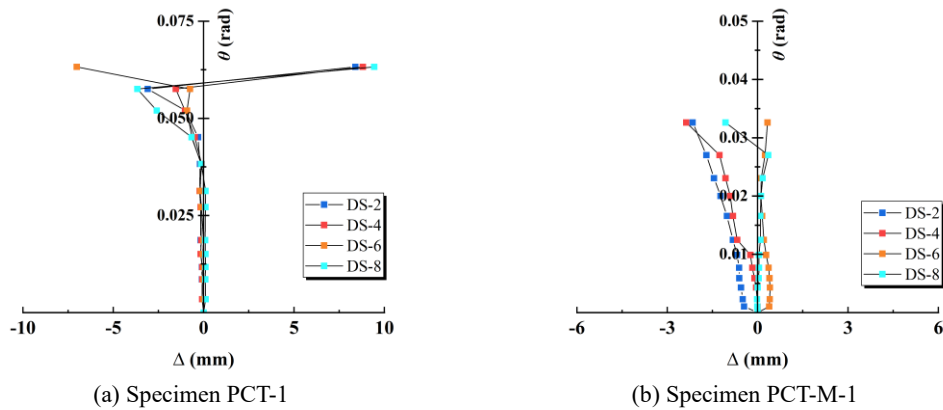


Fig. 10 Strain increment-torsion angle curves

3.5 Displacement increment curve

As shown in Fig. 11(a), the overall vertical displacement of the specimen under pure torsional load remains essentially constant. This stability is attributed to the nature of pure torsional behaviour, which generates minimal vertical displacement. This observation aligns with the trend of actual prestressing force increment changes. As illustrated in Fig. 11(b) - (d), the compression zone of the specimen is located on the northwest side, while the tension zone is on the southeast side. This distribution is primarily due to the radial component of the applied horizontal torsional load, which induces a bending moment in the northwest direction on the horizontal joint. In conjunction with the applied horizontal load, the horizontal joint experiences a bending moment directed northward. Consequently, during the loading process, the resultant bending moment on the horizontal joint remains directed northwest, resulting in compression on the northwest side and tension on the southeast side, which corresponds with the actual force mode of the specimen.



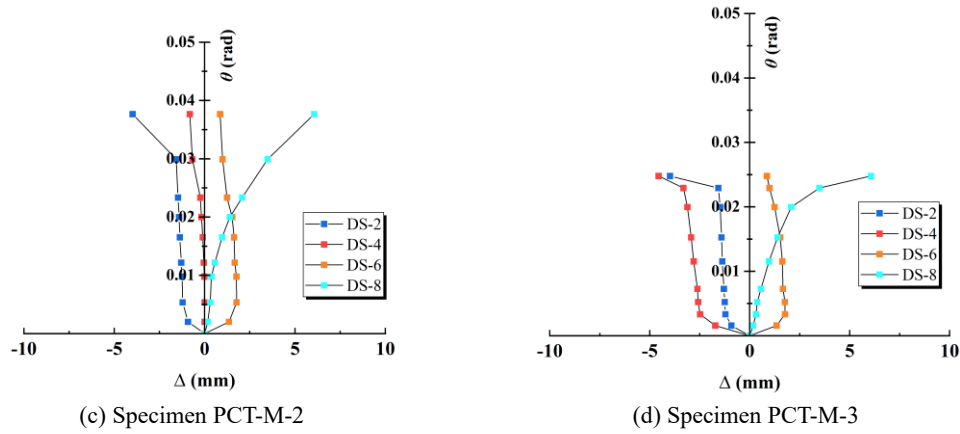


Fig. 11 Displacement increment-torsion angle curves

4 FE modelling

ABAQUS [22] was employed to develop a FE model of the tested specimens. Both geometric and material nonlinearities were considered in the numerical simulation. Additionally, the specimen and loading procedure are reasonably simplified to enhance computational efficiency.

4.1 Material properties

The Mander [23] constitutive curve was adopted for the elastic-plastic constitutive model of concrete. The specimen was equipped with double-layer of stirrups, and only the confinement effect of the outer layer stirrups on the concrete material was considered. The fracture energy of the concrete was calculated according to *CEB-FIP MC90* [24] and the tensile strength of the concrete was determined according to *GB 50010* [25]. An ideal elastic constitutive model was employed for the foundation and bolts, while an ideal elastic-plastic constitutive model was used for the steel reinforcement and prestressed tendons, as specified in *BS EN 1992-1-1* [26]. Additionally, the expansion coefficients of 10^{-5} are applied to the prestressed tendons.

4.2 Contact interaction

The "Embedded Region" was used to simulate the interaction between the steel reinforcement and the concrete. The surface-to-surface contact was adopted at four interfaces: the interfaces between the transition and upper segment, the interface between the lower segment and the foundation, the interface between the foundation and the ground, and the interface of the horizontal joint. The tangential behaviour was defined by the penalty friction formulation with a friction coefficient of 0.5 and the normal behaviour is defined by "hard contact". A reference point RP-1 was established

at the mid-height of the loading connection device and was coupled with the bolt holes on the upper part of the transition segment. A reference point RP-2 was established at the upper surface of the loading connection device for the prestressed tendon anchor and coupled with the upper position of prestressed tendons and the upper surface of the transition segment. The loading connection device was reduced to a truss element, and coupled with the transition segment. The bottom position of prestressed tendons was coupled with the reserved holes in the foundation.

4.3 Meshing and boundary conditions

The eight-node linear solid element (C3D8R) was used for the concrete structures. The steel reinforcement, bolts, the connection device and prestressed tendon only bear axial loads, so the two-node linear truss element (T3D2) was employed for the simulation. Shell elements (S4R) was used for the foundation. After the mesh sensitivity analysis with consideration of the balance of mesh density and time cost, the final mesh size of the transition segment, ground and the foundation was selected as 120×120 mm, the mesh size of the concrete structure and steel reinforcement in the upper segment and lower segment were selected as 30×30 mm. The ground part and the loading connection device were considered as rigid body. Meanwhile, the three translational degrees of freedom U1, U2 and U3 and three rotational degrees of freedom UR1, UR2 and UR3 were fixed for the bolt hole areas of the foundation part and the ground part. This transverse load and displacement load were applied at reference point RP-1 and RP-4. The specific mesh and constraint settings for each component are shown in Fig. 12.

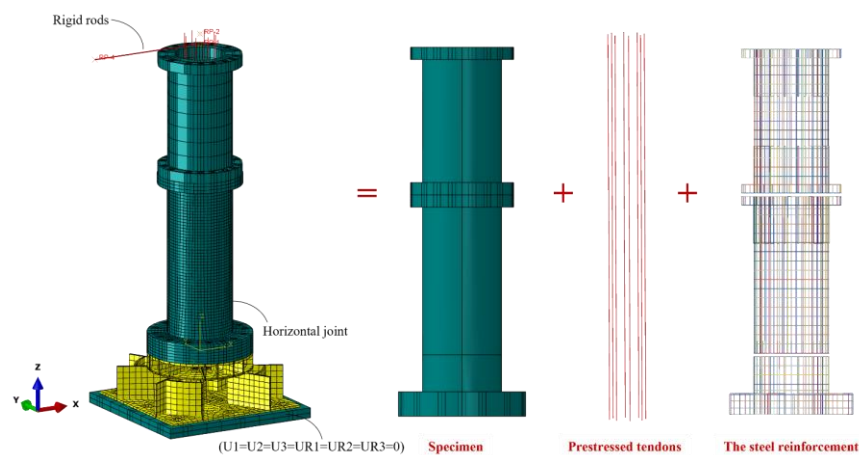


Fig. 12 FE model

4.4 Model validation

The comparison of the torsion angle-torque curves obtained from experiments and numerical simulations are illustrated in Fig. 13. It is evident that the experimental and FE curves for all specimens exhibit good agreement. Table 2 summarises the torsional capacities determined from each test and the corresponding FE modelling. The ultimate capacities obtained by FE model are differ from the experimental results by less than 5%. the FE model will be employed for a parametric study, as discussed in the subsequent section.

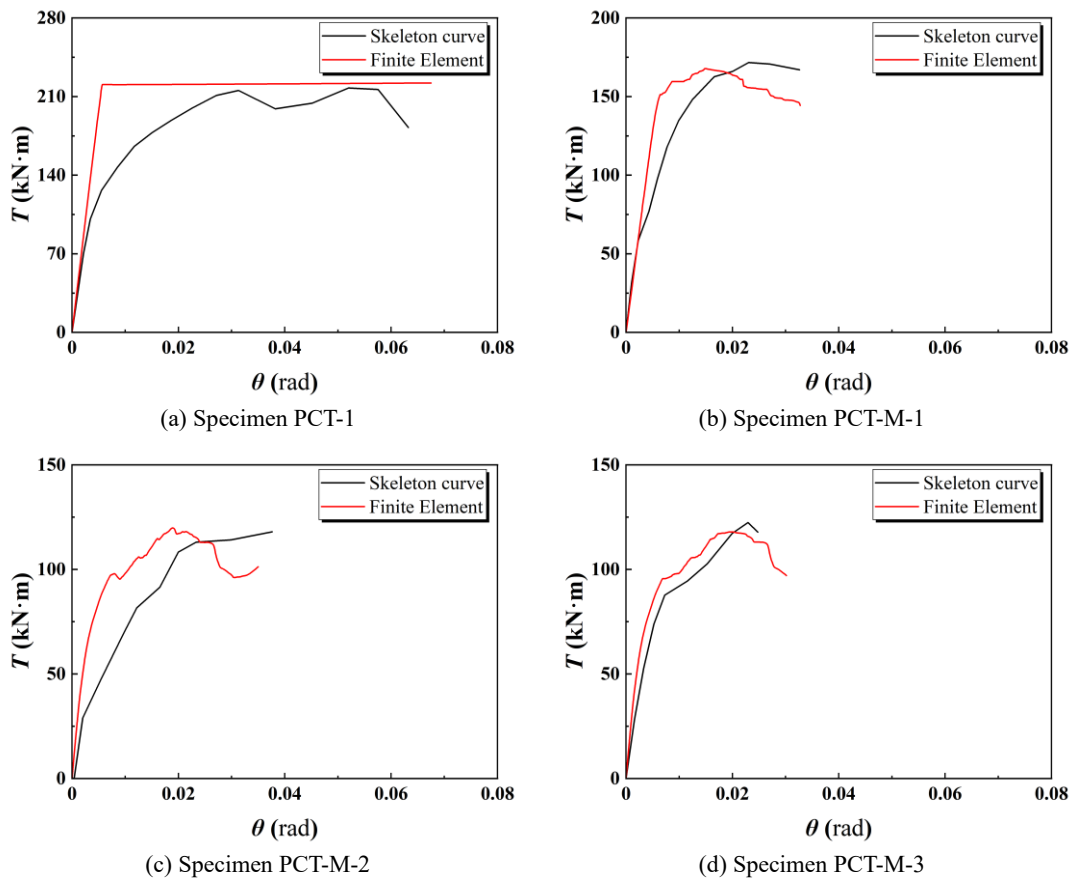


Fig. 13 Comparison of skeleton curves between FE model and test

Table 2 Comparison of the torsional capacity between FE model and test (kN·m)

Specimens	PCT-1	PCT-M-1	PCT-M-2	PCT-M-3
Test	217.44	171.4	119.76	122.4
FE	219.96	167.7	121.92	118.2
FE/Test	1.01	0.97	1.02	0.96

5 Parametric study

To investigate the factors influencing the torsional capacity of horizontal joints in prestressed

concrete towers for wind turbines, a parametric study was conducted. In this analysis, using the FE verification model as a basis, the minimum ratio of reinforcement of the actual prestressed concrete towers for wind turbines was applied.

5.1 Effects of material properties

Concrete strength and the yielding strength of the steel reinforcement were identified as potential factors influencing. In the parametric study, the concrete strength grade (f_c) was set to ranging from C60 to C90, while the yielding stress of the steel reinforcement (f_y) varied from 235 MPa to 435 MPa; both are typical material strengths used in the practical applications. The results are presented in Fig. 14, which shows that the torsional capacity of the horizontal joint is not significantly affected by the strength of concrete and steel reinforcement when they are within the range of practical applications. This may be attributed to the fact that the torsional capacity of horizontal joints is primarily related to friction, which is determined solely by the axial load for the specimens. Therefore, these influences will not be included in the calculation method.

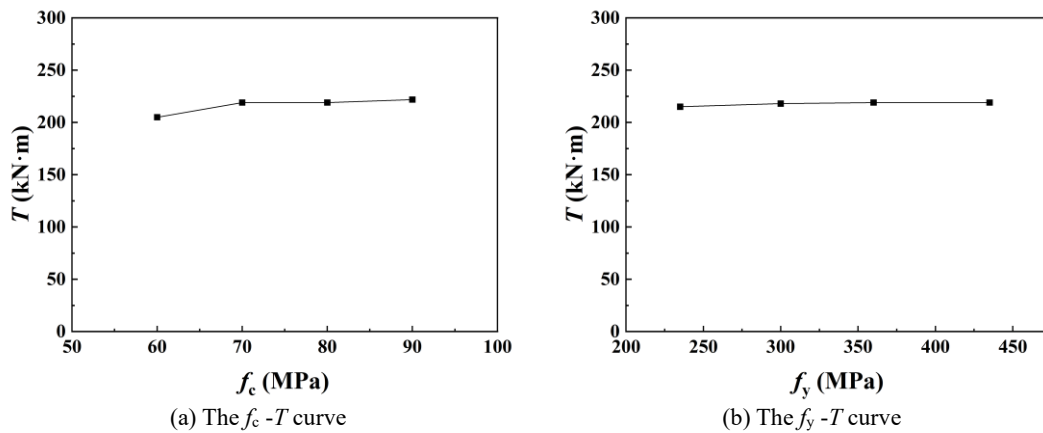


Fig. 14 Effects of the material strength on torsional capacity of horizontal joints

5.2 Effects of cross-sectional dimensions

The cross-sectional dimensions of horizontal joints in actual prestressed concrete towers for wind turbines with a circular profile are primarily determined by the radius and wall thickness. In the parametric study, the radius (r) ranging from 325 mm to 900 mm, and wall thicknesses (t) varied from 50 mm to 90 mm were employed to investigate the effect of the cross-sectional dimensions on the torsional capacity of the horizontal joint. The results indicate that the torsional capacity of horizontal joints increases with the cross-sectional diameter but is not significantly affected by wall

thickness, as shown in Fig. 15. This may be attributed to the fact that the torsional capacity of the horizontal joint is determined by the axial load and the lever arm. Changes in the section radius directly influence the size of the lever arm; however, since the hybrid tower is a thin-walled structure, variations in wall thickness have a minimal impact on the lever arm.

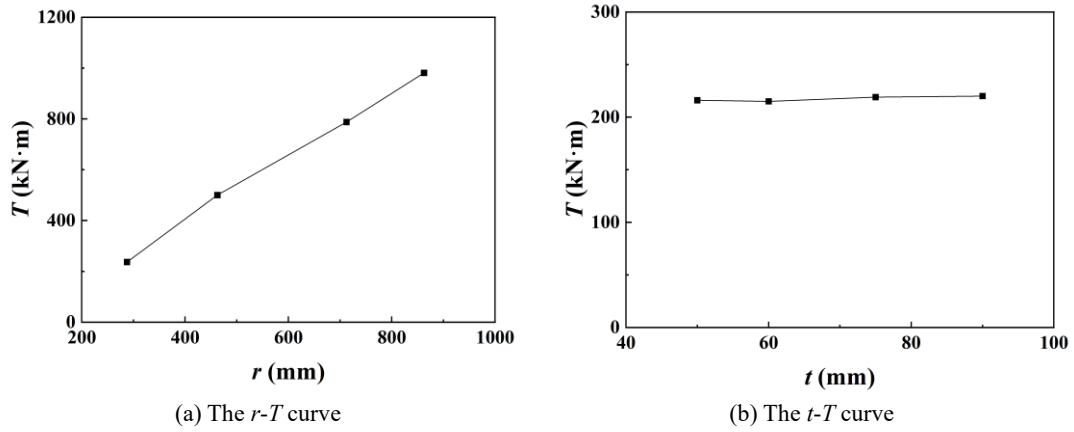


Fig. 15 Effects of cross-sectional dimensions on torsional capacity of horizontal joints

5.3 Effects of prestressing force

To ensure that the prestressing force in the section aligns with the actual hybrid tower structure, the prestressing force applied to the specimen is controlled by a precompression ratio that comprehensively considers the geometric characteristics and material properties of the cross-section dimensions [27]. The precompression ratio of the specimen is calculated using Eq. (1).

$$\alpha_{py} = \frac{F_{py}}{f_{ck} \cdot A} \quad (1)$$

where α_{py} represents the precompression ratio of the specimens; F_{py} denotes the prestressing force sum of the specimen; f_{ck} indicates the characteristic compressive strength of concrete; A represents the full cross-sectional area of the horizontal joint in the specimen.

Fig. 16 illustrates the effect of different prestressing forces on the torsional capacity of the horizontal joint. Within the precompression ratio range of 0.1 to 0.5, torsional capacity increases with the pre-compression ratio. As the pre-compression ratio rises, the torsional capacity of the horizontal joint exhibits a significant yielding phenomenon. This is primarily because, at relatively high precompression ratios, the failure mode of the specimen shifts from slip failure for horizontal joint to torsional failure for reinforced concrete. In the current wind power industry, the precompression ratio range is typically set between 0.15 to 0.35 to meet the fatigue requirements of

hybrid tower structures. Therefore, the range is applied to subsequent analyses.

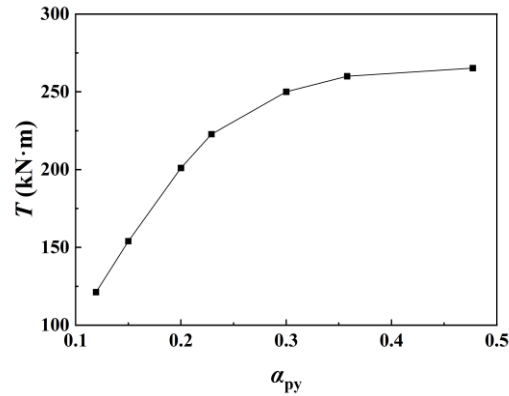


Fig. 16 Effects of the precompression ratio on torsional capacity of horizontal joints

5.4 Effects of transverse loads

When the horizontal joint in prestressed concrete towers reaches its ultimate torsional capacity, cracks begin to form at the joint. A transverse load was applied to the specimen, which induces horizontal shear forces and bending moments in the horizontal joint. When the bending moment is sufficiently large, the phenomenon of horizontal joints cracking will be observed. A portion of the friction force at the joint counteracts the horizontal shear force, defining the transverse load at which the tensile stress in the tension side of the joint reaches zero as the critical cracking load (F_0) of specimen. As the transverse load is applied, it has been observed that the increase in load leads to a more significant decrease in the torsional capacity of the horizontal joint, as shown in Fig. 17. Furthermore, the transverse load directly influences the torsional capacity of the horizontal joints.

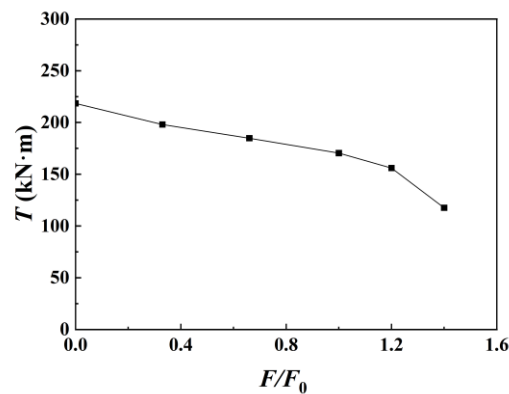


Fig. 17 Effects of the horizontal external load on torsional capacity of horizontal joints

In summary, the transverse load is a critical factor affecting the torsional capacity of horizontal joints in prestressed concrete towers for wind turbines. Meanwhile, the torsional capacity of

horizontal joints is primarily attributed to interfacial friction force, which is directly related to the interfacial friction coefficient, the vertical resultant force, and the lever arm. Therefore, understanding the specific impact of the prestressing force within the horizontal joint and the transverse load on the stress mode of the horizontal joint is crucial for calculating the torsional capacity. Once the cross-sectional shape, material properties, and prestressing force are established, the transverse load becomes the sole factor affecting the torsional capacity of horizontal joints in prestressed concrete towers for wind turbines.

6 Design methods

6.1 DIN EN 1992-1-1

The design rules are stated in clause 6.3.2 in *DIN EN 1992-1-1* [17]. The design method in generally the thin-walled tubular structures design is applied. When the horizontal joint is not cracked, the friction force is used only to resist torsion at the horizontal joint. When the horizontal joint cracking, the torsional capacity is obtained by applying the equivalent principle of maximum shear stress of the thin-wall close cross-section bar.

When uncracking of horizontal joints, the design method of torsional capacity (T) of the horizontal joints is calculated by Eq. (2).

$$T = \mu F_N r \quad (2)$$

When cracking of horizontal joint, the torsional capacity (T) of horizontal joints is calculated according to Eq. (3).

$$T = \mu F_N \frac{t}{3} \quad (3)$$

where μ represents the friction coefficient of horizontal joints; F_N indicates the vertical resultant of horizontal joints is calculated by Eq. (4).

$$F_N = F_Z + F_{py} \quad (4)$$

where F_Z represents the gravity load of specimens; F_{py} indicates the prestressing force.

The geometric characteristics of the cross-section of horizontal joints are shown in Fig. 18.

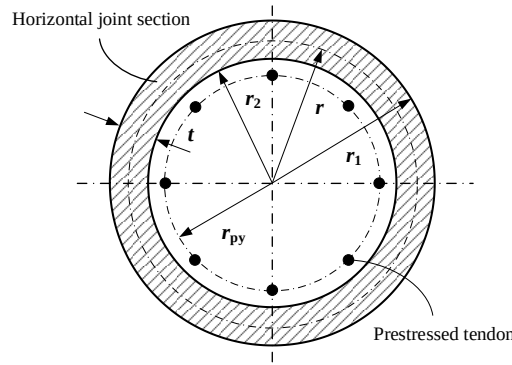


Fig. 18 The geometric properties of horizontal joints

The wall thickness t of the horizontal joint is calculated according to Eq. (5).

$$t = r_1 - r_2 \quad (5)$$

where r_1 denotes the external radius of the horizontal joint; r_2 indicates the internal radius of horizontal joints; r represents the radius of the centreline of horizontal joints is calculated using Eq. (6).

$$r = \frac{r_1 + r_2}{2} \quad (6)$$

6.2 T-CEC 5008

The design rules are stated in clause 6.2.3 in T-CEC 5008 [21]. The design method generally applied for thin-walled circular hollow section beam design is utilized under conditions of shear force and torsion. This calculation principle assumes that the combined shear and torsional effects at the transverse joint are resisted by friction force. The calculation method for the torsional capacity (T) of horizontal joints is provided in Eq. (7).

$$2V + \frac{T}{r} \leq \mu F_N \quad (7)$$

where V represents the shear force in the horizontal joints. The remaining symbols are defined in the same way as in section 6.2.

6.3 Proposed design method

To explore the structural failure mechanism and torsional capacity of the horizontal joint in prestressed concrete towers for wind turbines. Based on the FE parametric analysis, theoretical analysis of the structural failure mechanism of horizontal joints is conducted.

6.3.1 Pure torsional condition

The torsional capacity of horizontal joints is provided by the frictional force. Under pure torsion conditions, the stress state of the horizontal joints is shown in Fig. 19.

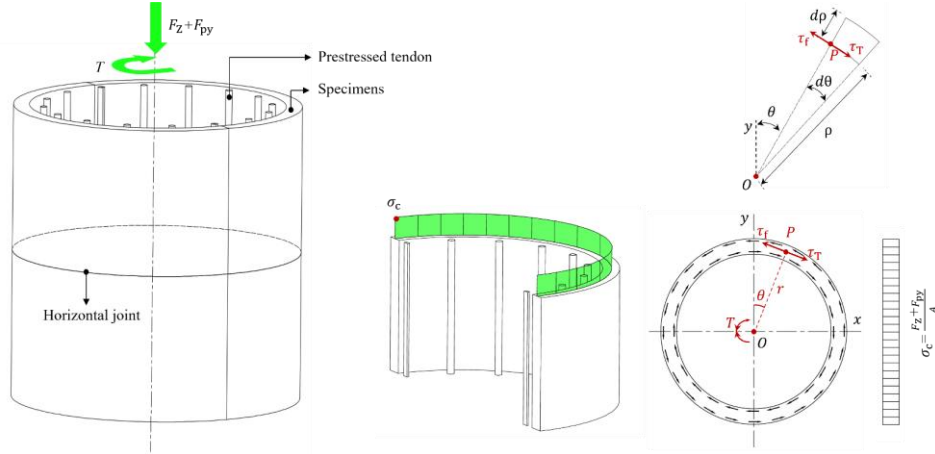


Fig. 19 Schematic diagram of the failure mode during pure torsional load of horizontal joints

The torsional capacity of the horizontal joint is equal to the frictional force moment provided by the interface. The torsional capacity of the horizontal joint of the prestressed concrete tower under pure torsion load conditions is calculated using Eq. (8).

$$T_0 = \mu F_N r \quad (8)$$

6.3.2 Bending-torsional condition

Before the horizontal joint cracks, the friction at the joint resists the combined effects of the torsional shear stress and the horizontal shear force. The stress state of the specimen at the horizontal joint at the point of critical cracking is illustrated in Fig. 20.

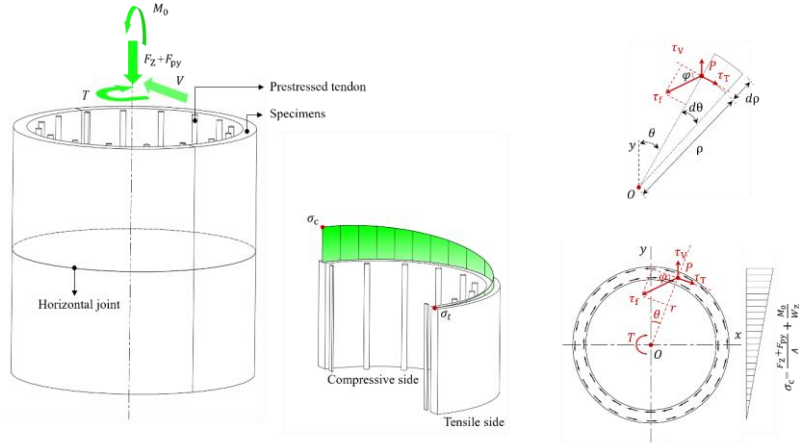


Fig. 20 Schematic diagram of the failure mode during the critical cracking state of horizontal joints

The stresses on the tension side of horizontal joints are calculated by Eq.(9), respectively. The stress at the edge of the concrete on the tension side of horizontal joints is exactly zero, which is denoted as $\sigma_t = 0$.

$$\sigma_t = \frac{F_z + F_{py}}{A} - \frac{M_0}{W_z} \quad (9)$$

where A is the horizontal joint area is calculated according to Eq. (10); W_z is the section modulus calculated by Eq. (11); M_0 is the critical bending moment for critical horizontal joints

cracking phase calculated by Eq. (12).

$$A = \pi(r_1^2 - r_2^2) \quad (10)$$

$$W_z = \frac{\pi r_1^3}{4} \left(1 - \left(\frac{r_2}{r_1} \right)^4 \right) \quad (11)$$

$$M_0 = \frac{F_z + F_{py}}{A} \cdot \frac{\pi r_1^3}{4} \left(1 - \left(\frac{r_2}{r_1} \right)^4 \right) \quad (12)$$

$$F_0 = \frac{M_0}{H} \quad (13)$$

where H represents the distance from the horizontal joint to the transverse load. F_0 indicates the critical shear force for critical horizontal joints cracking phase.

As the bending moment increases, tensile stress develops on the tension side of the horizontal joint, leading to cracking. This analysis considers only the levelling effect of the epoxy resin in the horizontal joints of actual prestressed concrete towers for wind turbines. Consequently, the horizontal joint cannot achieve complete shear flow. The stress state of the specimen when it reaches its torsional capacity is illustrated in Fig. 21.

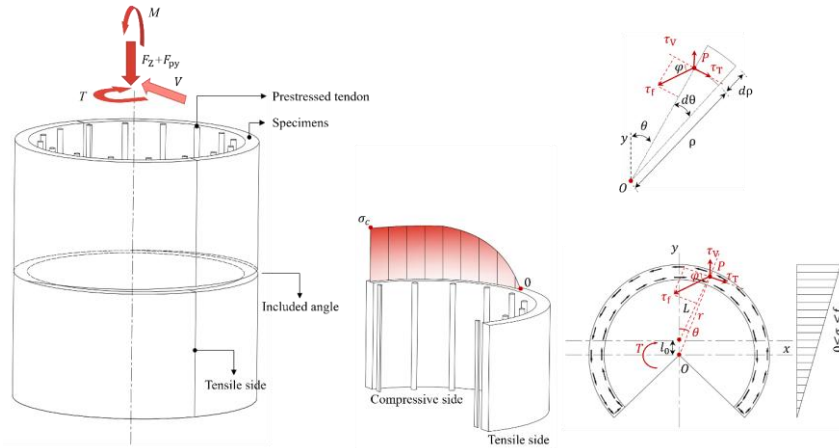


Fig. 21 Calculation diagram for the torsional capacity of horizontal joints

The calculation of the torsional capacity of horizontal joints is calculated according to Eq. (14):

$$T = \varphi \mu F_N r \quad (14)$$

where φ represents the reduction coefficient.

6.3.3 Determination of coefficients in design formula

When the horizontal joint of the FE model is influenced by bending moments but remains uncracked, the friction in the horizontal joint will bear shear forces and bending moments. At uncracking phase, the bending moments only cause stress redistribution in the horizontal joint, and the friction in the horizontal joint mainly resists the resultant force of the torsional shear stress and the shear stress. The functional relationship between the reduction coefficient φ and the shear force F/F_0 before the

horizontal joint cracking, as shown in Fig. 22. The functional relationship between the reduction coefficient φ and the shear force F/F_0 in the horizontal joint is calculated according to Eq. (15).

$$\varphi = -0.47 \left(\frac{F}{F_0} \right)^{1.6} + 1 ; (0 \leq \frac{F}{F_0} \leq 1) \quad (15)$$

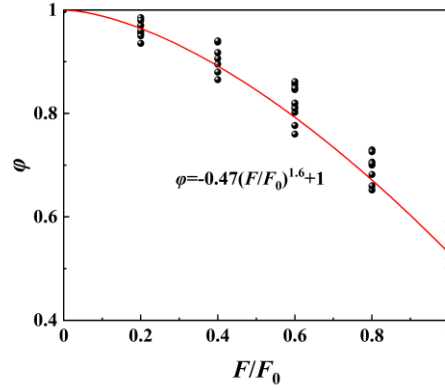


Fig. 22 Reduction coefficient of torsional capacity during horizontal joint uncracking

At post-cracking phase, the contact area of horizontal joints decreases, which causes the shift of the torsional centre, leading to smaller lever arm of the resultant frictional force for torsion. Concurrently, the part of the friction resists the shear force. The functional relationship between the reduction coefficient φ and the cracking ratio v of horizontal joints, as well as the shear load F/F_0 , after the horizontal joint cracking of the specimen and reach the torsional capacity, as shown in Fig. 23. The reduction coefficient and the cracking ratio v of horizontal joints is calculated by Eqs. (16) - (17).

$$v = 0.6 \left(\left(\frac{M}{M_0} \right) - 1 \right)^{0.5} \quad (16)$$

$$\varphi = 0.14 \left(\left(\frac{F}{F_0} \right) - 0.95 \right)^{-0.2} + 0.25(v + 0.95)^{-1.5} \quad (17)$$

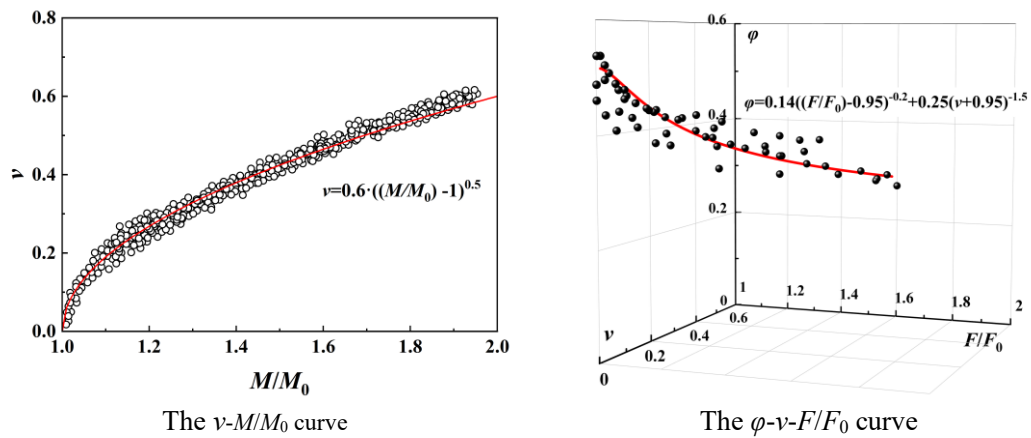


Fig. 23 Reduction coefficient of torsional capacity during horizontal joints cracking

6.3.4 Design procedure

The calculation process for determining the torsional capacity of prestressed concrete towers for

wind turbines is primarily divided into three steps, as outlined in Eqs. (18) - (22).

Step 1: Based on the cross-sectional dimensions, prestressing force, and self-weight data, the critical bending moment and shear force in the horizontal joint during the critical cracking phase are calculated by Eqs. (18) - (19).

$$M_0 = \frac{F_z + F_{py}}{A} \cdot \frac{\pi r_2^3}{4} \left(1 - \left(\frac{r_1}{r_2} \right)^4 \right) \quad (18)$$

$$F_0 = \frac{M_0}{H} \quad (19)$$

Step 2: The occurrence of cracking in the section is assessed by comparing the bending moment acting on the section with the critical bending moment. If the section bending moment exceeds the critical bending moment, the cracking ratio is calculated by Eq. (20).

$$v = 0.6 \left(\frac{M}{M_0} - 1 \right)^{0.5} ; (1 < \frac{M}{M_0} < 2) \quad (20)$$

Step 3: The reduction coefficient is calculated based on the shear force and the cracking ratio affecting the horizontal joint by Eq. (21). Finally, the proposed design method for the torsional capacity (T) of the horizontal joints is calculated according to Eq. (22).

$$\varphi = \begin{cases} -0.47 \left(\frac{F}{F_0} \right)^{1.6} + 1 ; (0 \leq \frac{F}{F_0} \leq 1) \\ 0.14 \left(\left(\frac{F}{F_0} \right) - 0.95 \right)^{-0.2} + 0.25(v + 0.95)^{-1.5} ; (1 < \frac{F}{F_0} < 2) \end{cases} \quad (21)$$

$$T = \varphi \mu F_N r \quad (22)$$

7 Result comparison and discussions

A comparison of the predicted torsional capacities of horizontal joints using existing and proposed design methods was conducted against the test and numerical results. Table 3 summarises the experimental torsional capacity (T_0) and the predicted capacities (T_1 for *DIN EN 1992-1-1* [17], T_2 for *T-CEC 5008* [21], T_3 for the proposed design method of this study).

Table 3 Comparison of calculation results for the torsional capacity (kN·m)

Specimen number	T_0	<i>DIN EN 1992-1-1</i> [17]		<i>T-CEC 5008</i> [21]		This study	
		T_1	T_1/T_0	T_2	T_2/T_0	T_3	T_3/T_0
PCT-1	217.44	218	1	218	1	219.6	1.01
PCT-M-1	171.4	224.39	1.3	180.8	1.05	124	0.72
PCT-M-2	119.76	19.37	0.16	162.3	1.35	74.3	0.62
PCT-M-3	122.4	23.6	0.19	161.9	1.32	72.3	0.59

Furthermore, Fig. 24 illustrates the relationship between the torsional capacity from the FE parametric analysis and those calculated using different codes herein.

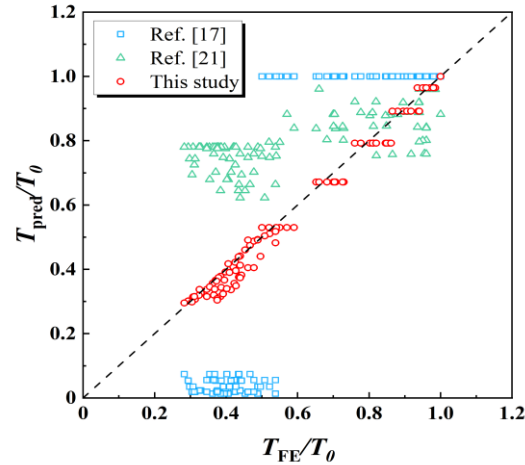


Fig. 24 Comparison between predicted results and simulated results

The comparisons clearly indicate that when the horizontal joints have cracks, the prediction of *T-CEC 5008* [21] was found to be unsafe, and of the predictions by *DIN EN 1992-1-1* [17] is over-conservative. However, when the horizontal joints have no crack, the predictions by *DIN EN 1992-1-1* [17] was on the unsafe side. It is clearly shown that the design method for the existing codes cannot accurately and reliably assess the torsional capacity of horizontal joints under the influence of bending moment and shear force, which was also reported by literature [19-20]. Table 4 summarises the comparison of design method based on all FE results. The new design method provides rather accurate and consistent predictions with torsional capacity a mean value of 0.96 and coefficient of variation (CoV) of 0.06.

Table 4 Comparison of design method between predicted results and simulated results

Design method	Mean	CoV
<i>DIN EN 1992-1-1</i> [17]	0.65	1.06
<i>T-CEC 5008</i> [21]	1.51	0.49
This study	0.96	0.06

8 Conclusions

A comprehensive experimental and numerical study on the torsional behaviour of horizontal joints in prestressed concrete towers for wind turbines is presented in this paper. The following conclusions can be drawn: (1) Under cyclic loading, the experimental program found that the torsional capacity of horizontal joints is primarily provided by the frictional force generated at the horizontal joints due to the prestressing force. Additionally, the transverse load significantly affects the failure range of the specimen at the horizontal joint. (2) The mechanism and parametric analysis were conducted based on the validated FE models of specimens. The precompression ratio and the

cross-sectional radius of the prestressed concrete towers for wind turbines significantly influence the torsional capacity of the horizontal joints. When the load on the cross-section is established, the sensitive parameters affecting the torsional capacity of horizontal joints include the bending moments and shear forces. (3) The accuracy and applicability of the calculation methods for torsional capacity in existing codes and literature were evaluated. The existing design methods were found to be either unsafe or overly conservative because they disregard the influence of bending moments, shear forces, and the cracking of horizontal joints. (4) Based on the results of the theoretical analysis of cross-sectional stress distribution patterns and FE parameter studies, a new practical calculation method was proposed that can significantly improve the design accuracy and efficiency for the torsional capacities of horizontal joints in prestressed concrete towers for wind turbines. This method takes into account the influence of bending moments and shear forces, with the mean value and CoV of the experimental-to-predicted ratios being 0.96 and 0.06, respectively.

Author contributions

Ji-Ke Tan: Conceptualization, Methodology, Investigation and Review & Editing. Wan-Lang Zheng: Investigation and Writing-Original Draft. Yong-Sen Lan: Supervision. Mei-Ni Su: Supervision and Review & Editing. Yun-Hui Zhang: Investigation. Hai Cao: Supervision.

Declaration of interest

No conflict of interest, financial or otherwise.

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